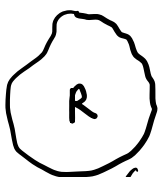


Lecture 19 Green's Theorem Cont'd

Recap:



C : simple closed curve

$\vec{F} = F_x \vec{i} + F_y \vec{j}$ is C^1 at every point in R

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA.$$

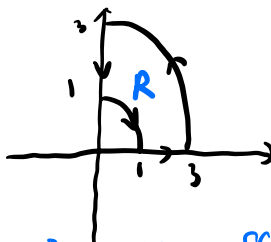
$$\nabla \times \vec{F} = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k} : \quad \oint_C F_x dx + F_y dy = \iint_R \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA.$$

Line integral $\rightarrow$ double integral.

Example. $\vec{F} = -y^2 \vec{i} + xy \vec{j}$ and C is a multi-segment path shown below.

Find $\oint_C \vec{F} \cdot d\vec{r}$

$$\begin{aligned} \text{Sol. } \nabla \times \vec{F} &= \left(\frac{\partial(xy)}{\partial x} - \frac{\partial(-y^2)}{\partial y} \right) \vec{k} \\ &= 3y \vec{k}. \end{aligned}$$



$$\text{By the Green's Theorem, } \oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA = \iint_R 3y \, dA.$$

The region R can be represented in polar coordinates as

$$1 < r \leq 3, \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

$$y = r \sin \theta \quad dA = r \, dr \, d\theta$$

$$\begin{aligned} \iint_R 3y \, dA &= \int_0^{\frac{\pi}{2}} \int_1^3 3r^2 \sin \theta \, dr \, d\theta \\ &= \left(\int_0^{\frac{\pi}{2}} \sin \theta \, d\theta \right) \left(\int_1^3 3r^2 \, dr \right) \\ &= [-\cos \theta]_0^{\frac{\pi}{2}} \cdot [r^3]_1^3 \\ &= 26. \end{aligned}$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 26.$$

Double integral $\rightarrow$ line integral.

$$\iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA = \oint \vec{F} \cdot d\vec{r}.$$

$$\text{If } (\nabla \times \vec{F}) \cdot \vec{k} = C_0, \text{ Area of } R = \iint_R dA = \frac{\oint \vec{F} \cdot d\vec{r}}{C_0} = \frac{\oint_C F_x dx + F_y dy}{C_0}$$

$$\text{E.g. } \vec{F} = \langle 0, x \rangle, \quad \vec{F} = \langle -y, 0 \rangle, \quad \vec{F} = \langle -\frac{1}{2}y, \frac{1}{2}x \rangle.$$

$$\nabla \times \vec{F} = \frac{\partial x}{\partial x} \vec{k} = \vec{k}$$

$$\text{Area of } R = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C (x \, dy - y \, dx).$$

Example (Area of ellipse) Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\text{Sol. } \vec{r}(t) = \langle a \cos t, b \sin t \rangle \quad 0 \leq t \leq 2\pi.$$

$$dx = -a \sin t \, dt \quad dy = b \cos t \, dt$$

$$\begin{aligned} x \, dy - y \, dx &= ab \cos^2 t \, dt - b \sin t (-a \sin t \, dt) \\ &= ab \, dt \end{aligned}$$

$$\text{Area} = \frac{1}{2} \oint_C (x \, dy - y \, dx) = \frac{1}{2} \int_0^{2\pi} ab \, dt = \pi ab.$$

Beyond computation

Let C be a very tiny simple closed curve enclosing a tiny region R .

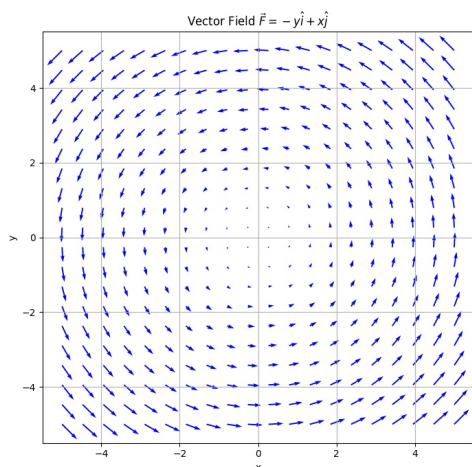


$(\nabla \times \vec{F}) \cdot \vec{k}$ can be regarded as a constant inside R

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA \approx (\nabla \times \vec{F}) \cdot \vec{k} (\text{area of } R)$$

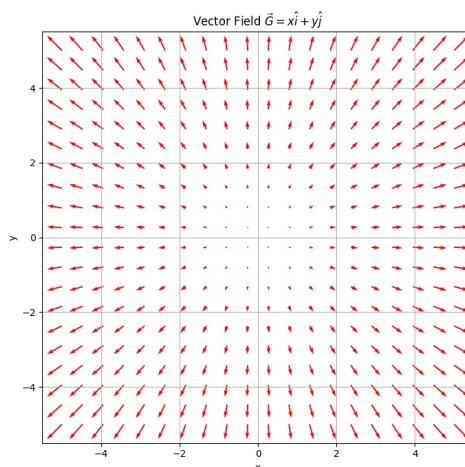
$$(\nabla \times \vec{F}) \cdot \vec{k} \approx \frac{\oint_C \vec{F} \cdot d\vec{r}}{\text{area of } R} = \frac{\oint_C \vec{F} \cdot \vec{r}'(t) \, dt}{\text{area of } R}$$

$\nabla \times \vec{F}$ is an indicator of curliness of $\vec{F}$



$$\vec{F} = -y\hat{i} + x\hat{j}$$

$$(\nabla \times \vec{F}) \cdot \vec{k} = 2$$



$$\vec{G} = x\hat{i} + y\hat{j}$$

$$(\nabla \times \vec{G}) \cdot \vec{k} = 0$$

Limitations of the Green's Theorem

$$\vec{F} = -\frac{y}{x^2+y^2}\hat{i} + \frac{x}{x^2+y^2}\hat{j}$$

$$\nabla \times \vec{F} = \vec{0} \quad \iint_R (\nabla \times \vec{F}) \cdot \vec{k} dA = 0$$

on unit circle C : $\oint_C \vec{F} \cdot d\vec{r} = 2\pi$,

Γ is a circle centered at $(3,0)$ with radius 2, then one can check that $\oint_{\Gamma} \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} dA = \iint_R 0 dA = 0$.

Winding number of a closed curve

$\oint_C \vec{F} \cdot d\vec{r} = 2\pi \times \text{number of times } C \text{ travels around the origin}$
counter-clockwise

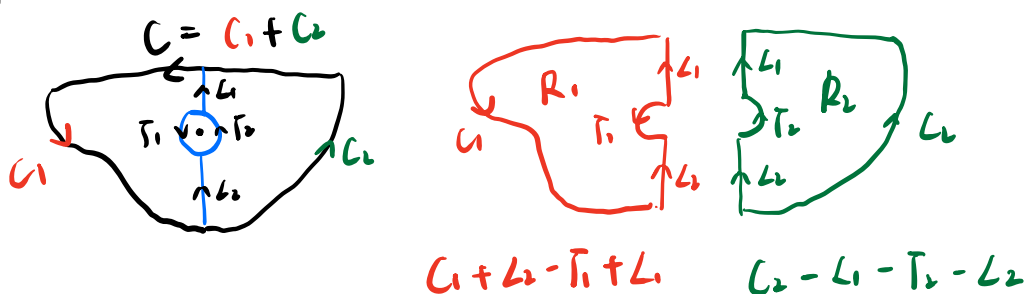
$\frac{1}{2\pi} \oint_C \vec{F} \cdot d\vec{r} = \frac{1}{2\pi} \oint \frac{-y dx + x dy}{x^2 + y^2}$ is called the winding number of C

Circles centered at origin

radius ε , $\Gamma_\varepsilon: \vec{r}(t) = (\varepsilon \cos t) \vec{i} + (\varepsilon \sin t) \vec{j} \quad 0 \leq t \leq 2\pi$.

$$\oint_{\Gamma_\varepsilon} \frac{-y dx + x dy}{x^2 + y^2} = \int_0^{2\pi} \frac{-\varepsilon \sin t d(\varepsilon \cos t) + \varepsilon \cos t d(\varepsilon \sin t)}{\varepsilon^2} = \int_0^{2\pi} dt = 2\pi$$

Simple closed curves enclosing the origin.



$$\oint_C \vec{F} \cdot d\vec{r} = \oint_{C_1} \vec{F} \cdot d\vec{r} + \oint_{C_2} \vec{F} \cdot d\vec{r}.$$

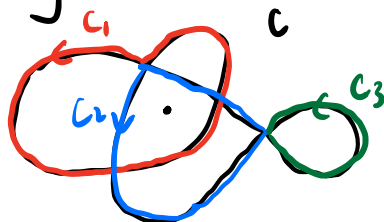
$$2\pi = \oint_{\Gamma} \vec{F} \cdot d\vec{r} = \oint_{\Gamma_1} \vec{F} \cdot d\vec{r} + \oint_{\Gamma_2} \vec{F} \cdot d\vec{r}$$

$$\textcircled{a} \oint_{C_1} \vec{F} \cdot d\vec{r} + \oint_{C_2} \vec{F} \cdot d\vec{r} - \oint_{\Gamma_1} \vec{F} \cdot d\vec{r} + \oint_{L_1} \vec{F} \cdot d\vec{r} = \iint_{R_1} (\nabla \times \vec{F}) \cdot \vec{e} dA = 0,$$

$$\textcircled{b} \oint_{C_2} \vec{F} \cdot d\vec{r} - \oint_{C_1} \vec{F} \cdot d\vec{r} - \oint_{\Gamma_2} \vec{F} \cdot d\vec{r} - \oint_{L_2} \vec{F} \cdot d\vec{r} = \iint_{R_2} (\nabla \times \vec{F}) \cdot \vec{e} dA = 0.$$

$$\stackrel{\textcircled{a} + \textcircled{b}}{\Rightarrow} \oint_C \vec{F} \cdot d\vec{r} = \oint_{\Gamma} \vec{F} \cdot d\vec{r} = 2\pi.$$

Self-intersecting curves.



$$\oint_C \vec{F} \cdot d\vec{r} = \oint_{C_1} \vec{F} \cdot d\vec{r} + \oint_{C_2} \vec{F} \cdot d\vec{r} + \oint_{C_3} \vec{F} \cdot d\vec{r} = 4\pi.$$